Reg. No. | | | | | | | | | | | [13EC65/12EC65]

SIXTH SEMESTER B.Tech/B.E UNIVERSITY EXAMINATION, JUNE – 2016

POWER ELECTRONICS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1. a. What is Power electronics? With neat circuit diagram and waveforms, explain the types of power electronic circuits. **08**

b. Explain control characteristics features of the following power devices: **06**

- **i.** GTO
- **ii.** MCT
- **iii.** SITH

2. a. Give the construction, static characteristics and applications of IGBT. **08**

b. A power BJT connected as a switch has $V_{CE}(sat) = 1.2v$ and β in the range 8 to 20. If Vcc = 200v, ODF = 6 and Rc = 18Ω. Compute **06**

- **i.** The Value of the base current required to ensure the transistor is in saturation
- **ii.** Forced β

3. a. Define the following terms with respect to SCR **06**

- **i.** Latching current
- **ii.** Holding current
- **iii.** Turn-on time
- **iv.** Turn-off time

b. With the help of neat circuit diagram and waveforms, explain the UJT-triggering circuit. **08**

4. a. Explain Single-phase Semi converter with a neat circuit diagram. **06**

b. What is a dual converter? Explain its operation with a neat circuit diagram. **08**

5. a. With a neat diagrams. Waveforms and equation discuss ON-Off control and phase control of AC voltage controllers. **10**

b. An AC voltage controller has a resistive load of R=10Ω and rms input voltage is Vs=120V, 60Hz. The thyristor switch is on for n=25 cycle and off for M=75cycles Determine **04**

- **i.** The rms out put Voltage
- **ii.** Power factor

6.	**a.**	What is commutation? Explain complementary Commutation with relevant circuit diagram and waveforms.	**08**
	b.	With a neat circuit diagram and waveform explain Resonant pulse commutation	**06**
7.	**a.**	Explain the principle of Operation of a step=up chopper.	**06**
	b.	With a neat circuit diagram and Quadrant Operation, explain class E chopper.	**08**
8.	**a.**	With the help of circuit diagram, explain Variable DC-link inverter and give Some advantages.	**06**
	b.	What is PWM? What are the Various PWM techniques? How do they differ from each other?	**08**